Cloud Foundation: AI-Assisted IaC



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AI-Assisted Migration: Usage and Disclaimer

Sharing of capabilities, not a formal delivery commitment

Partner training | Huawei Cloud IaC for cloud foundation

Central Note

This session presents the AI-assisted capabilities of Huawei Cloud for building cloud foundation with Infrastructure as Code. It does not process real customer data nor constitute a formal delivery plan, security or compliance commitment, or product feature promise.

1. Data and Privacy

Security / Compliance / Protection

- No real customer data is collected or processed
- Use simulated, test, or anonymized data
- Do not upload accounts, passwords, keys, certificates, or data

2. Usage Limits

Reference / Human Review

- AI supports planning, analysis, scripts, and risk assessment
- It does not replace design, expert review, testing, or decisions
- Review plans, scripts, and configurations before the project

3. Customer Authorization

Authorization / Approval

- The customer decides whether to use AI in real projects
- Without authorization, do not enter customer information into AI
- Respect security, compliance, confidentiality, and approvals

4. Not Recommended Scenarios

Prudent Evaluation / Manual Process

- Evaluate or avoid highly sensitive security scenarios
- Classified systems, financial core, defense, or energy
- Use manual migration under strict sovereignty or AI prohibition

AI-assisted migration improves efficiency; it does not replace human review or customer authorization. Real implementation must be based on customer confirmation, security, compliance, and project situation.

The Cloud Foundation Challenge



Why Cloud Foundation?

Every migration needs a solid base: networking, security, IAM, monitoring. Without foundation: manual console clicks, drift. With IaC: reproducible, version-controlled, auditable. Landing zone in 1 day with AI.



Key Challenges

Service mapping: AWS/Azure -> Huawei Cloud. IAM: translate roles and policies. Networking: VPC, subnets, routing. Security: security groups, KMS, WAF. Compliance: data residency, audit logging. Cost: right-sizing.



AI-Assisted Approach

AI generates Terraform from requirements. AI maps AWS/Azure services to Huawei Cloud. Natural language cloud ops via MCP. AI validates templates. All infrastructure as code, no console clicks.

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MCP Servers & IaC Workflow

Huawei Cloud MCP



hcloud MCP: 40+ structured tools. list_servers, list_vpcs, obs_ls, list_cce_clusters. Natural language: list all ECS in cn-north-4. 2 seconds vs 5 min manual console.

Terraform MCP



Registry: search_modules, get_module_details. Provider: get_latest_provider_version. Workspace management: create, update, list runs. Plan/apply monitoring from AI. Private registry for enterprise.

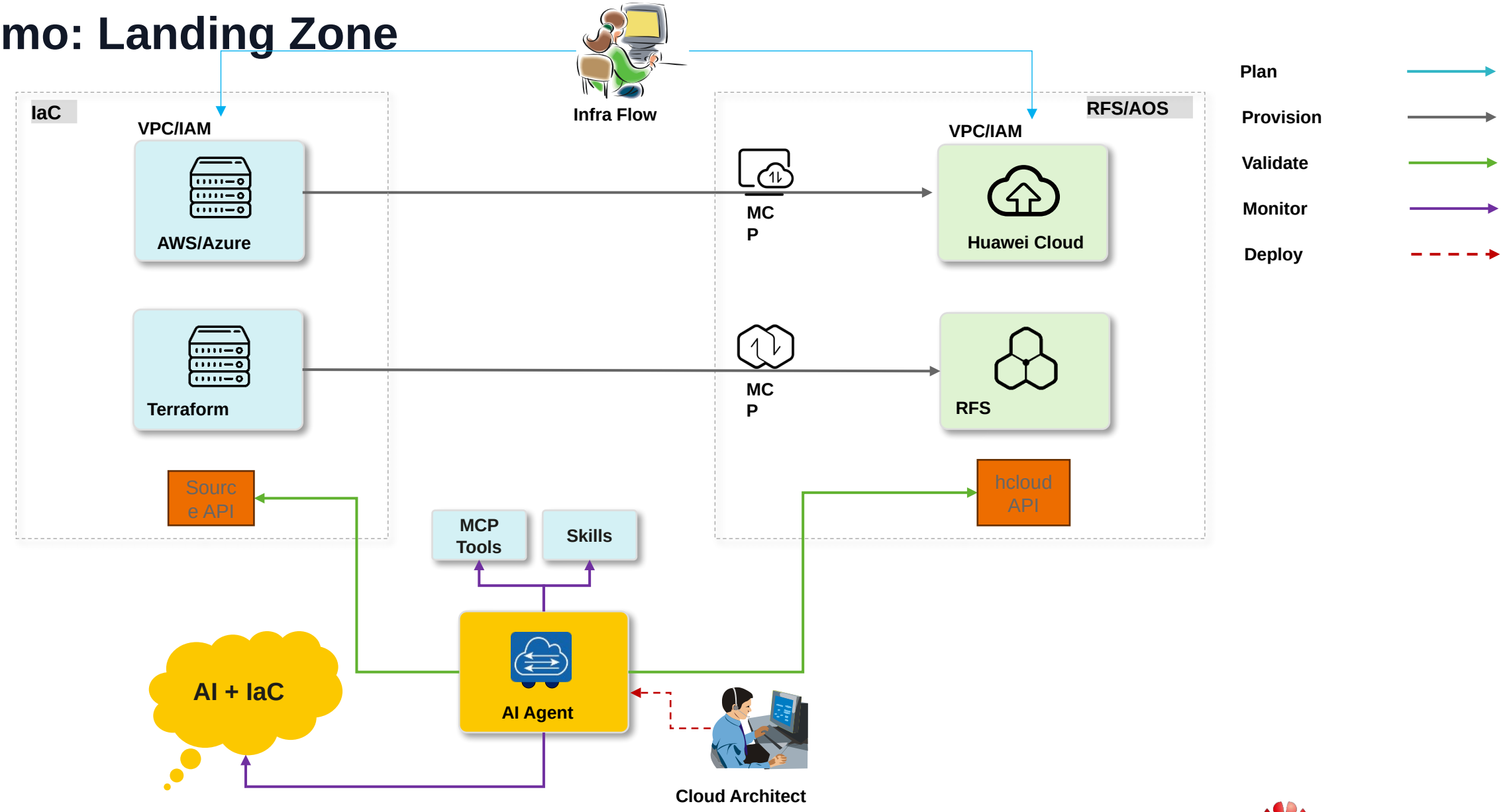
Landing Zone Design



Hub VPC: NAT, DNS, VPN/Direct Connect. Spoke VPCs: dev, staging, prod. Multi-region: active-active or pilot-light. Typical: 1 hub + 3 spoke + 2 regions. ~500 lines HCL, 1 day with AI.

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Demo: Landing Zone



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AWS/Azure to Huawei Cloud Mapping

Category	AWS/Azure	Huawei Cloud	Tool
Compute	EC2/VM	ECS	Terraform
Containers	EKS/AKS	CCE	Terraform
Storage	EBS/Disk	EVS	Terraform
Object	S3/Blob	OBS	Terraform
Network	VPC/VNet	VPC	Terraform
Security	SG/NSG	SG	Terraform
IAM	IAM/Entra	IAM	Terraform
Monitoring	CloudWatch	CES	Manual
Key Mgmt	KMS/KeyVault	KMS	Terraform
Migration	VM/DB	SMS/DRS	Terraform
IaC	CloudFormation	RFS/AOS	HCL

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Monitoring & Security

CES

Cloud Eye Metrics

Features:
ECS, CCE, OBS, RDS metrics
Custom metrics and dashboards
Alarm rules and notifications
Auto-scaling triggers

Usage:
Monitor all foundation resources
Alert on threshold breach

AOM

App Operations

Features:
Application performance monitoring
Log collection and search
Trace analysis
Component topology

Usage:
Full-stack observability
CI/CD integration

LTS

Log Tank Service

Features:
Centralized log management
Structured log search
Log alarm rules
Log ingestion from ECS/CCE

Usage:
Audit trail via CTS
Security analysis

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AI-Assisted Benefits

Operational Efficiency

AI generates Terraform from requirements.
AI maps AWS/Azure to Huawei Cloud.
Natural language cloud ops via MCP.
AI validates templates before deploy.
Landing zone: 1 day vs 1 week.

Standardization & Risk

All infrastructure as code, no console clicks.
Consistent across environments.
Drift detection: RFS compares state.
Audit trail: CTS logs all changes.
Backup + DR from day 1.

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Resources & Next Steps

Resources

github.com/binrogithub/
1-3-Cloud-Adoption-Skills/
tree/main/AI-Assist-Migration/
Cloud-Foundation

Next Steps

1. Service mapping (Day 1): AI maps source.
2. Landing zone (Day 2-3): AI generates VPC, IAM.
3. IaC templates (Day 4-6): RFS/AOS.
4. Deploy (Day 7): AI validates + hcloud CLI.
5. Operations (Day 8-10): monitoring, backup.

Contact

Huawei Cloud LATAM SMB team
for landing zone reference architecture

IaC + MCP + Terraform for reproducible,
version-controlled infrastructure.

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BENEFITS



Operational Efficiency

AI generates Terraform from requirements. AI maps AWS/Azure services to Huawei Cloud. Natural language cloud operations via MCP. AI validates templates before deployment.



Cost Optimization

Landing zone: 1 day with AI vs 1 week manual. Template deploy: 8 min vs 2 hours. Right-sizing: AI recommends optimal flavors. No over-provisioning from day 1.



Risk Reduction

Drift detection: RFS compares state. Audit trail: CTS logs all changes. Backup + DR from day 1. Security baseline enforced via IaC.

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